Perception as Entanglement: Chris Fields

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so i'm i chose a pretty provocative title today saying forget about emergence and i'm going to try to convince you that it's actually a good idea to forget about emergence and to forget about reductionism too and see what we're left with okay so i've got to point it to nope doesn't work at all worked a few minutes ago there we go so we're going to start with a little poll how many of you have ever heard of quantum computing raise your hand not know anything about it just have you heard of it so it looks like at least a fair number of you have how many of you have ever heard that quantum computing is a lot faster than ordinary computing so you know whoever figures out quantum computing is going to be able to wreck all of our financial transactions and all of that put our banks in trouble and sort out the defense department and so on okay so today's questions are what's the difference between classical computing and quantum computing and they were going to segue from that into what does this tell us about perception uh human perception and perception in general and then we're going to talk about what happens when we just forget about these concepts of emergence and reduction what are we left with in studying perception so let's talk uh about quantum computing but first i want to put one assumption on the table i'm going to assume that basically i'm going to assume that quantum theory is a correct description of the world but what i'm really assuming is that two symmetries of quantum theory are correct descriptions of the world and i'll talk about each of those symmetries as we go through this and i suspect that we can get quantum theory from those two symmetries and nothing else uh no other assumptions so we're going with a different clicker now okay well as long as i have the laser pointer on fire i can i can click with that that's no problem yeah this still works brilliant okay just means i have to wear glasses so keep in mind though that this is an assumption and if it turns out that these symmetries are violated for real in the world then every everything i'm going to tell you today is wrong uh because it's all based on on the assumption that the symmetries of quantum mechanics actually hold so this is a a very deep as as don was mentioning in his case a very deep empirical assumption is that the universe has the particular symmetries we'll be talking about let's talk about quantum computing well now actually no methods of advancing the slides work let's go a little bit farther okay there we go all right now it should work okay great thank you very much for the rescue okay um so let's talk about quantum computing quantum computing is really very simple and this is the simplest possible representation there are many many representations of quantum computing they're all provably identical so it doesn't matter which one we talk about so we're going to talk about this one which is simple the simple version of quantum computing is you start with a question and a unitary process is executed and you get an answer this is called the hamiltonian oracle view it's the model of quantum computing that david deutsch used to design the quantum turing machine model it's provably complete so anything you can do with the quantum computer designed any way in the world you can do this way so that the heart of this is this thing i've called a unitary process and unitary processes are actually the heart of quantum mechanics so this is what a unitary process is it's a process that you can reverse and get the identity so the way to think about this dynamically is a unitary process is a process that is perfectly time symmetric so this is symmetry number one of quantum theory quantum theory is perfectly time symmetric evolution is unitary and i would claim that if you think hard about what time symmetry means you will understand quantum theory in an intuitive way but you have to really think about time symmetry because it's something that's radically unlike what our world is typically and we'll see some of the ways in which it's radically unlike what our world typically is so time symmetry in particular means causal symmetry so we have a question we have an answer and in a symmetric unitary process you can think of it as a combination of a forward process that goes from the question to the answer and a reverse process that starts with the answering goes back to the question in fact there's a beautiful series of papers by giuseppe castiglioni castignoli sorry uh in italy and he has shown that if you think of a quantum computation in this way as starting simultaneously at the question and the answer and working toward the middle and ending with the two processes kit you know like the golden spike in the railway then you can completely understand the speed up of quantum computation over classical computation and the way he puts it is that a quantum computation always knows half of the answer up front because it's working at the same time from the answer backwards in time and there's a very nice treatment of this in ruth castner's book the transactional interpretation of quantum mechanics which adds some other stuff that i think is probably not quite right but she does a nice treatment of time symmetry in this forward and reverse way however there's a problem with this picture and you all have probably already sorted out what the problem is the problem is that these words question and answer are not time symmetric words in our vocabulary right you ask the question first and then you get an answer it's not we don't think of that in a time symmetric way the problem is that we've taken some informational concepts question and answer and we've defined quantum computing by mixing these up with unitary evolution which is a purely physical concept that has to do with time symmetry and so we've gotta we've got to to say how these match up to have a theory that actually means anything now technically the last picture i showed you is completely meaningless so that let's fix that by making explicit that the unitary process acts on some physical states some states in the world that are subject to dynamics but question and answer are semantic concepts so what we've done i want to emphasize the we here some observer has interpreted this physical stuff going on in terms of a question and an answer now that's a semantic process that's giving something meaning so we've constructed a map a semantic map from something down here in dynamics world to something up here in semantics world and that's what allows us to talk about problem solving we interpret something going on in the world as a question and then we interpret something going on in the world as an answer and something or other has happened in the middle which we may or may not know how to interpret or how to measure anything else but we know how to do something very important we know how to say i'm going to ask a question and get an answer and then i'm going to use the answer to be in my next question and get another answer and get another answer so we know how to iterate this process and when you iterate this process you get a whole chain of events up here all of which are semantics that sort of right on top of this dynamics down here that you may have no characterization of at all but and here's the second symmetry of quantum theory your interpretation up here the semantics what you call things doesn't change the dynamics that's an it's a very important assumption because without that assumption we can't do science at all if what i call this lectern if i call this a blidget doesn't change the laws of physics if it did science would be impossible every language would lead to to not just an impression of different laws of physics but actual different behavior of the world so we have to assume that's not the case uh that's sort of the most rudimentary uh assumption of communicative objectivity that you can make that semantics doesn't change the physics that means there's no such thing as physical collapse so you've all heard of collapse of the wave function doesn't exist and i'll show you why and if you're not convinced by the end of the talk ask questions in the question period so what are these induced mappings up here from question one to question two to answer number two here that's just turning computation that's exactly what a turing machine is it's just turning one symbol into another where the symbols are semantic interpretations of something or other going on in the world so for example you know we can interpret some state of the world as that equation and do an operation called addition and get that representation and multiply and get that representation we're just manipulating symbols but this stuff down in the bottom all this physics that's going on in your laptop or your head or somewhere okay so what have we learned here we learned that classical computation is just a semantic interpretation of something going on in the world that's describable by quantum theory in particular this thing is a quantum computer your iphone is a quantum computer there are quantum computers that are carefully designed so that you don't realize it so that you can treat them as perfectly classical devices that behave in reproducible ways the thing is a quantum system i mean on the inside it's just a bunch of energy and stuff moving around okay uh forget about the difference between classical and quantum computation it doesn't exist at the physical level so here's an example right uh this thing is a bunch of energy and if i look at it at kind of a small scale i see that that's the inside of a central processor of a computer now when i was a graduate student we had a computer called a pdp9 in our lab is about the size of six or seven refrigerators if you open up the back doors it looked just like that but you had to go in there and move the wires around sometimes and in the 50s all programming was done by moving wires around so that right there was the user interface it was just you know twice as big as this area right here now we have user interfaces that look like that you know and i think i'm dealing with powerpoint and linux and you know a page viewer or something like that all of that's just a fiction like don talked about this morning it's just software but this stuff is just software too right we defined this this is something that we we configured the physical world to make it easy to manipulate for us we said that's a device but it's really just a user interface it's an interpretation of what's going on so we can generalize this and say any classical computation is a user interface or on a quantum computer and in fact all classical symbols of all kinds all of language is a user interface uh so we're done with the first part i want to emphasize that i've done nothing fancy here at all this is old stuff uh from textbooks this is just standard denotational semantics which is the techniques that one uses to define programming languages and it's been around forever uh it all rests on a paper that alfred tarski published in a philosophy journal philosophy and phenomenological research back in the 40s when he first developed the theory of how to do the semantics of formal languages expressed in logical notation so this is this is ancient standard stuff and i want to emphasize nowhere in this talk am i doing anything weird or new okay so what does this mean for so let's look at the usual picture here usual picture is something happens in and someone sees it or detects it somehow or other and they have an experience and maybe they do something maybe there's some behavior so this perceiver here is obviously a little bit concerned about what they saw so we want to look at this process so let's slow time down a little bit look at what happens okay so there's an event and then some information travels through space and it whacks the perceiver and it causes an experience and then the experience causes a response on the perceiver's part so this is all very classical so let's look at this again and think of think of how classical this is something happens in space and time information moves at a finite speed right the speed of light or less it causally impacts the perceiver and then the perceiver reacts now note that is not time symmetric i have not given you a quantum mechanical description this is as classical as the day is long so let's put this in quantum theory terms here's the quantum picture of the same thing there's an initial state of the world in which we have an event and a perceiver and the perceiver hasn't noticed the event yet that's just a snapshot of the world let's call that the initial state there's a process that acts on this entire system to produce some final state which consists of the perceiver their experience their reaction and if the event is still happening it's still there maybe it's gone away but this whole thing turns into that whole thing by a unitary process and remember these things are processes that are time symmetric and they're independent of how we describe them two symmetries so what's happening here physically physically we say that some event coupled to some observer evolves through time to some event coupled to some observer just at a different time so i'm writing these things coupled together that's what that cross with the little circle means to say this is all one thing it's all evolving together in time okay and it works both ways now this is what's physically going on what's the result the result is that this event gets classically correlated with the experience that's what we used to say by cause but there's a correlation now between the event and the experience or in information theoretic terms the observer now has information about the event so there's a word for this in quantum theory it's entanglement there's a physical process going on in an entangled state with a bunch of components and that process as a result classically correlates two components of that state so how many of you have heard of entanglement most of you how many of you have heard that entanglement has something to do with faster than light communication okay you're all in good company because of course einstein himself gave you that interpretation back in his famous paper in the 30s and if you think about entanglement this way you will never understand it ever here's the definition entangled in quantum theory just means not separable it's all it means and not separable means this it means you cannot equate the state of an entangled state with a bunch of components you cannot equate it to some mixture of the states of the individual you can't say this thing is a and b interacting that's wrong so read not equals as if i think about it this way i will get it wrong every time okay so what entanglement means is one physical system with one state so whenever you hear uh i have two entangled photons right that's wrong that's crazy you don't have two entangled photons you've got one physical system with two photons as components they're not interactive at all they're just an entangled system let's think about this let's look at the standard nicolas geisen type experiment and i picked gyson because he did this kind of experiment first over a separation of about 10 kilometers about 10 years ago in geneva and the most recent paper i've seen which was just uh 1310 so that's this month uh was from a japanese group we've done the same experiment over 300 kilometers so next time someone says quantum mechanics that's just about little bitty stuff okay remember when you thought cells were little bitty things and then someone said oh you've got meter long cells in your legs you said oh white cells are big like that so quantum systems are big like 300 kilometers long so quantum mechanics is not a microscopic phenomenon it's a macroscopic phenomenon 300 kilometers is big so what happens in an experiment like this you've got a source and the source emits an entangled state of something so maybe it's an entangled state of two photons two electrons you know in reality most of these experiments are done with photons so the photons come out and they're coupled in the spin coordinates so one of them spin up and one spin down typically easy to make okay they move apart and they're two observers and the observers are always called alice and bob because it's more fun than calling them a and b okay so allison alice and bob have detectors and they're separated in space by a long way the photons if you think about it if the photons are coming from this source toward alice and bob they're going to get to alice and bob in the same amount of time or less than it would take for alice or bob to send a message to each other and they'll get there in half the time for alice to send a message to bob and bob to send the message back okay hence the faster than light confusion so what's going on here the photons come out of the source and everything in sight is entangled right the whole system moves as one and the light moves a little bit more it's still entangled a little bit more still entangled and then suddenly alice and bob get a signal they say up down okay now they know they've observed an entangled state but it was entangled all along right what else and bob know is absolutely irrelevant to the physics that's the symmetry what we call it doesn't matter so here's the message of entanglement this is what quantum mechanics has been trying to tell us for 80 years and we're like this okay we don't want to hear it it's trying to tell us there are no boundaries the universe is an entangled system there's just one thing and it evolves through time in quotation marks now as one single system now physicists hate this why i mean this is what quantum mechanics is telling us why is it a problem it's a problem because it makes it very very hard to say what an experiment is right physicist wants to say okay we're going to manipulate the world somehow or other and see what happens but if the whole thing moves as one no manipulation going on and you know physicists are little bitty things and the rest of the universe is a really big thing so if anything at all is going on in experiments nature is manipulating us right us manipulating nature that's nonsense it's crazy it's silly right nature manipulates us when we do an experiment and physicists call this conspiracy that nature is only going to reveal what she wants to well of course but there's a lovely article by uh gerard stuhoft a couple of months ago again on the archive it's not published yet um and in this article he basically says conspiracy is just the way it is gang get used to it and he has a nobel prize so he's much much smarter than i am but that's the message in any rate and it's good that that it's being said by people who are taken seriously so let's do gysen's experiment again keeping in mind that everything is one quantum state so it's entangled still entangled still entangled still entangled and this business of knowledge is irrelevant alice and bob don't know they're entangled for a long time but they're entangled through the entire process so here's our result entanglement is how classical information is transferred from one place to another so entanglement is what implements it's the one mechanism we have within quantum mechanics to implement and it implements it all the time ah it says okay this thing over here is entangled with that thing over there so we will exchange some classical information and note that that's relative it's always relative to the specification of a coordinate system and by the way this uh perceiver perceived boundary there's no boundaries in physics so that boundary is just semantics including my boundary with you your boundary with your chair all that's semantics okay so let's put this all together almost done with the second third of the talk so all reportable experiences the things we can talk about write down make a movie about store in a disk drive anything we can report put in a memory and expect to say the same all reportable experiences are included in classical information that's what classical information is stuff you can put it in memory all classical information is semantics it's just a user interface on what's dynamics physics whatever you want to call it we are classically correlated with the world so we can get information we can learn something by entanglement and the boundaries that separate the us from the world are just part of our user interface so i want to reference don here because he's been saying this for a long time we've at least come to it this way by a different route that's complementary to his there's an epistemic bottom line here oh it says all reportable knowledge all knowledge that we can tell someone about or write down somewhere or even just remember from one moment to the next even just remembering something from one moment to the next is making a semantic that your memory is the same thing that it was when you encoded that piece of information a moment ago you have any evidence for that no one does so don't say yes there is no evidence for that and there's no way to ever get any evidence for that so it's just an assumption that's why it's semantics so all reportable knowledge is both contextual depends on the specification of some coordinate system that lets us talk about entanglement and it's relational it's about my relation to something i'm entangled with so knowledge is of the form if x then y there's no such thing as knowledge of the form x blanket statement all knowledge is implication or so that's all nice philosophy so what well i'll argue that there are three so what's okay so here's so what number one this makes a powerful empirical prediction and this is in fact the prediction that quantum theory has been trying to get us to take seriously for 80 years it's entanglement is everywhere every experiment at any scale that's designed well enough should detect so it's no surprise that over the last five years we've been getting more and more and more evidence that you can only understand human cognition in quantum mechanical and there are now dozens of beautiful experiments but the simple one which you may have heard of is the pet fish experiment how many people have heard of the pet fish experiment okay pet fish experiment goes like this i say imagine a pet okay so everybody imagine a pet how many raise your hand if you imagine guppy nada forgetting about that question which i know you can't do imagine a fish just the prototypical fish how many of you imagine guppy one two all right so now imagine a pet fish how many guppies lots okay now classical probability theory tells us the probability of event a times the probability of event b is the probability of event a b combined well we just did an experiment where we had a really low probability of saying a pet is a guppy and a pretty low probability of saying a fish is a guppy and a really high probability of saying a pet fish is a guppy so we've all in this room just violated classical probability theory now it turns out that you can describe that kind of experiment beautifully using the mathematics of quantum theory and you've just created an entangled state and now they're now experiments over and over and over and over again showing this in different domains and the math is getting better and more concepts are being brought in like entanglement and people are getting braver to some extent to say look we're seeing quantum mechanical effects going on in human cognition so i would predict and and schrodinger could have predicted i think he did actually at least some of his writings that we'll see that everywhere you know 50 years we'll say classical probability that doesn't work anywhere so that's the empirical prediction there's a methodological prediction which is there aren't any parts so if you go around saying the whole is more than the sum of the parts well so you're not saying anything so the whole emergence paradigm is out the window forget about that but of course if you go around saying well we can explain everything in terms of what the parts are doing why not any parts so the whole reductionist paradigm is out the window too so we need to forget about most philosophy of science which is either emergent dust or reductionist it's just wrong i mean if quantum theory is right it's quantum theory is wrong maybe it's right but we know that quantum theory is right that whole kind of thinking is confused if we're going to understand experience then and we can't use emergence and we can't use reduction but none of us want to use reduction anyway but if we can't use emergence we've got to study entanglement and by this i mean don't just think about it and it's good to think about it but we need to do where we try to understand how human beings are actually coupled to the world in quantum mechanical terms we've got a good understanding of that in classical but we don't understand it in quantum mechanical terms at all and so there's a huge new frontier of experimental science and they're going to be experiments that are hard because what's nice about classical science is that you can reproducibly create the same conditions over and over and over again quantum theory tells you that's impossible you're never reproducibly creating the same conditions over and over again so you've got to be really creative to create conditions that you can pretend are the same and get away with it and when they're doing those experiments you know very near absolute zero in fiber optics and all of that they can pretend that they're reproducibly creating the same conditions and get away with it come up with pretty good evidence that they're getting away with it but it's harder in psychology but it's something we've got to figure out how to do so finally so what number three is kind of the moral or maybe spiritual but but i regard it as a moral point that observers are everywhere that's an observer you know the system that consists of my big toe and my glasses is an observer the black hole in the middle of the milky way is an observer uh so there are observers everywhere at every scale and most of their experiences are nothing whatsoever like ours so you have your experience think about milky way what's its experience like you're probably saying who knows well we don't know we don't have a clue but we can assume pretty clearly that it's nothing like our experience and i think what that gives us is a certain amount of humility to think that our experiential position in the universe is not privileged in the least there's nothing special about being us there are lots of other systems out there that have a rich experiential life they're nothing like we are and their opinions of things are probably radically different from ours with that i'm done questions thank you very much chris uh elizabeth rauscher and i have been talking about the possibility of putting a josephson junction in a new metal shield and then having a second identic virtually identical hopefully doesn't junction that then can be modulated as you would a a radio wave or a television broadcast and to see whether the under superconducting conditions whether this uh already entangled state of all observers would allow for information transfer that was uh could be considered to be instantaneous so that we would have we could do without radio magnetic transmissions i i would guess that you won't transfer classical information that way you'll transfer plenty of quantum information that way i know you want to transfer classical information could you make that distinction more clear like can we you transfer useful well i think that's a different dimension i mean they're at at a physical level all the information that we use is quantum information it's useful to the extent that we've survived and can do things but the information that we write down in memories and assume is going to stay the same so in a sense all the information that we think of as cultural is classical by definition now the question is what's the interpretive process that allows you to say that a piece of information is classical it involves making an assumption about its existence over time and it's an assumption that we don't understand very well and we understand something about how the brain executes the process of making that assumption how that you know i'm the same object that i was five minutes ago you've probably all assumed that but you know no evidence in the world that says that that's true it's something that we make up um so your your question goes to that core semantic issue of what does it mean to say that i've gotten a useful piece of classical information from point a to point b and there's a very familiar question in quantum computing because it's exactly the question of i define some state of a quantum computer as an input and then i wait a while go have a cup of coffee come back and the quantum computer's in another is that state that it's now in you know the two factors of some you know 10 to the 97th digit prime number well i can find out in that case by multiplying the answers together and see if it's what i define to be the initial but in most cases we can't do that we have no idea how to do the semantics of quantum processes to achieve a ever more refined many more observations occurring for a moment these answers and questions are meeting and being dissolved through fun processes in this evolution that uh our interest in being able to communicate instantaneously across space and perhaps any separation in time is actually an illusion and so we can actually travel in those space and time and commute and communicate perhaps across these artificial areas except if there's no such thing as time i have no idea what the word travel means we are um coming to the end of our time we can keep going if because we have a panel on which you are on coming up if uh if the people that are setting up the panel want to do that while you're talking if that you want to keep taking questions we can do that we can have other questions if we want all right thank you chris a question from a classical psychologist as opposed to a physicist then uh your little guppy example uh if i look at that from the point of view of let's say a social psychologist what you did was bias your respondents by contextualizing in the information that you provided is in your questions so in since you biased us toward a certain set of responses can you reinterpret that in entanglement yeah contextuality is almost another word for entanglement in actually the the results within quantum theory illustrate the effect of entanglement on supposedly independent experimental processes are actually called contextuality results in analog with what you see in psychology i suspect that a lot of classic psychological data can actually be reinterpreted in quantum theoretic terms in these sorts of effects many of which we understand to some extent mechanistically with in terms of what's going on in the brain we can say this is a case that actually illustrates quantum statistics in action i mean there wouldn't be contextuality if we lived in a in a perfectly classical world we have another question back here could you elaborate on the point you the observers are everywhere and what impact that has i didn't quite follow what that means in practical terms okay that's why i carry dwight as a moral point because we tend to think of ourselves and you know maybe mammals and maybe trees as observers we don't think of computers as observers or chairs or as observers or planets as observers for example but if you if you think of observation as information transfer uh and you think of information transfer in physical terms then it becomes very difficult uh what's different between information transfer in my case and information transfer and you know a thermostat's case to take the classic silly example that philosophers use to say well you know surely you can't be a pan psychist because then you'd think that thermostats made decisions in my opinion the right response to that say yeah and there's a there's a wonderful paper by christopher fuchs who's the guy who kind of invented quantum bayesianism he's a quantum computation quantum information guy he says in one in in in this paper uh with reference to something called the free will theorem you know does that does his theory actually imply that elementary particles are making decisions he says yeah it does that's just get used to it you